

James Hamilton

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EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems

Manhattan, NY

Bachelor of Science (BS) in Computer Science | GPA: 3.68

December 2026

EXPERIENCE

New York State EMS & Hospital Network (via Pace University)

New York, NY

Software Engineer Intern (Clinical AI & Full-Stack)

May 2025 - Present

- Architecting a live-triage EMS application to securely route patient data across the NY hospital network.
- Developing a decoupled architecture, building a native **Android (Java)** client to consume **AI** backend services.
- Designing backend AI logic to optimize emergency resource allocation under zero-fail constraints.
- Engineering real-time, data-compliant pipelines to synchronize critical healthcare metrics.

Arcampass

Remote

Software Engineer (Founding Team)

February 2025 – January 2026

- Built and deployed a production web platform using **Next.js**, **Express**, and **MongoDB**, supporting **200 monthly active** users and handling **1,000** API requests per day.
- Designed **MongoDB** schemas for user accounts, sessions, and transactional data. Added compound indexes on high traffic queries, reducing average read latency from **320 ms** to **140 ms**.
- Containerized backend with **Docker** and deployed to **AWS EC2** behind **NGINX**. Reduced deployment time from 12 minutes to 8 minutes and improved API latency to **180 ms** using **Redis** caching.

Pace University, Seidenberg School of CSIS

New York, NY

AI Research Assistant- LLM Code Automation

April 2025 – September 2025

- Developed a **Python** benchmarking framework to evaluate **5 LLMs** across **50** structured software engineering tasks.
- Measured code correctness, runtime failures, and security violations. Identified a **28 percent average** failure rate in authentication and input validation logic across models.
- Automated evaluation using unit tests and static analysis checks, reducing manual grading time by **60 percent** and improving result reproducibility.

BZ Marketing

Remote

Software Engineer Intern

May 2024 - September 2024

- Led frontend development of a **React** landing page redesign resulting in a SEO score increasing from **71** to **88**.
- Built an AI driven keyword generation feature using **OpenAI** and **Django**. Increased client campaign outreach by **20 percent** within **2 months** of deployment.
- Implemented **Redis TTL** caching and optimized **PostgreSQL** queries in the CRM dashboard. Reduced average page load time from **2.4 seconds** to **0.9 seconds** and cut third party API costs by **15 percent**.

PROJECTS

Master Parcel Data Engine - Next.js | Node.js | Go | PostgreSQL | AWS | Python | Redis | MongoDB

- Architected a real estate DaaS platform** featuring a hybrid **Go** and **Python (Pandas)** ETL pipeline that standardizes unstructured **MongoDB** market listings and municipal records (MapPLUTO, ACRIS) to the NYC BBL standard.
- Built a serverless Next.js/React frontend** powered by a secure **Node.js** backend to manage API routing, protect environment variables, and handle email-gated data streaming.
- Integrated a zero-cost architecture** using **Supabase (PostgreSQL)** for automated lead capture and CRM routing, alongside **Stripe** for seamless B2B monetization.

Parkidsy (Cross-Platform Web Application) - Next.js | Bun | Supabase (PostgreSQL) | TypeScript

- Architected a mobile-optimized **full-stack** web application wrapped inside **native iOS** and **Android WebViews**.
- Leveraged the **Bun** runtime to build high-performance, low-latency backend API routes.
- Engineered a relational **Supabase (PostgreSQL)** schema for secure authentication and real-time data storage.

INDEPENDENT INITIATIVES

Commuter App Design & Launch Challenge with MTA & NJ Transit Leaders (Finalist)

November 2024.

- Designed core features using Kotlin/Swift and Golang with RESTful APIs, implementing user authentication, offline caching, and Microsoft Azure database integration for seamless and secure functionality.

TECHNICAL SKILLS

- Languages:** Python, TypeScript, Golang, JavaScript, C#, HTML, CSS, SQL
- Frameworks:** React.js, Next.js, Express.js, Node.js, Deno, .NET
- DevOps/Cloud:** Docker, AWS EC2, S3, Route 53, NGINX, Azure, CI/CD
- Databases:** MongoDB, PostgreSQL, Firebase, Redis, Supabase